COVID-19 Information Portal

Software Testing and Quality Assurance (STQA) Mini Project

Candidate / Developer: Omkar Ahirrao

Course: Software Testing & QA (STQA)

Architecture: HTML5, CSS3, JS/React 19, Node.js / PHP, PostgreSQL / MySQL

Execution Status: 47 Test Cases Executed (100% Pass Rate)

1. Executive Quality Assurance Summary

This official test document reports the systematic verification and quality assessment of the **COVID-19 Dynamic Information & Awareness Portal**. Developed according to academic STQA practical curricula (Test Scenarios for Web Applications, Test Case Design using Boundary Value Analysis and Equivalence Class Partitioning, Regular Expression Validation, and Automated Test Execution), the portal features dual-tier input validation, security hardening against XSS and SQL Injection, and live database persistence with ACID guarantees.

47	47	5	42ms
TOTAL TEST CASES	TESTS PASSED (100%)	DEFECTS TRIAGED & FIXED	CLOUD DB LATENCY

2. Regular Expression Master Specifications

Dual-tier regular expressions are evaluated both client-side for real-time visual feedback and server-side to guarantee rejection of malicious or malformed payloads:

INPUT FIELD	FORMAL REGULAR EXPRESSION	CONSTRAINT DESCRIPTION	VALID SAMPLE	INVALID SAMPLES
Full Name	`^[A-Za-z ]{2,50}$`	2 to 50 characters, letters and single spaces only	`Omkar Ahirrao`	`O`, `Omkar123`, `Omk@r`, `51 chars`
Email Address	`^[^\s@]+@[^\s@]+\.[^\s@]+$`	RFC standard email: local-part, @, domain, dot, TLD	`omkar@gmail.com`	`omkar@gmail`, `omkar@.com`, `omkar@gmail.com`
Mobile Number	`^[6-9][0-9]{9}$`	10 digits Indian format starting with 6, 7, 8, or 9	`9876543210`	`98765432` (8 dig), `98765432100` (11 dig), `5876543210`
Password	`^(?=.*[A-Za-z])(?=.*\d).{8,}$`	Minimum 8 characters with at least 1 letter and 1 digit	`Omkar123`	`Omk12` (<8), `OmkarAhirrao` (no num), `12345678` (no let)
Comment Text	Trimmed: `2 to 500 chars`	Non-empty, length bounded, safe HTML encoding	`Stay safe everyone!`	Empty string, whitespace only, >500 characters
Feedback Msg	Trimmed: `5 to 1000 chars`	Minimum 5 chars for actionable review	`Helpful symptoms tool.`	Empty string, <5 characters

3. Table 1: Functional Test Cases Matrix (35 Cases)

Covers User Registration, Authentication, Regular Expression Assertions, Symptoms Risk Calculation, Forum Comments, and Admin Management:

TC ID	MODULE	TYPE	TEST DESCRIPTION & OBJECTIVE	TEST DATA / REGEX INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
`TC_FUNC_001`	Register	Positive	Verify user registration with valid credentials	Name: `Omkar Ahirrao`, Email: `omkar@gmail.com`	Account registered; HTTP 201 Created	Account created in Supabase DB	PASS
`TC_FUNC_002`	Email	Negative	Verify regex rejection of email missing TLD	`omkar@gmail`	Rejected: invalid email format	Rejected by regex /^[^\s@]+@[^\s@]+\.[^\s@]+$/	PASS
`TC_FUNC_003`	Email	Negative	Verify regex rejection of email missing domain	`omkar@.com`	Rejected: domain name required	Rejected as expected	PASS
`TC_FUNC_004`	Email	Negative	Verify regex rejection of email with embedded space	`omkar @gmail.com`	Rejected: spaces disallowed	Regex failed on space character	PASS
`TC_FUNC_005`	Mobile	Positive	Verify acceptance of valid 10-digit mobile	`9876543210`	Accepted with green checkmark	Regex /^[6-9][0-9]{9}$/ matched	PASS
`TC_FUNC_006`	Mobile	Boundary	Verify rejection of 8-digit mobile (below boundary)	`98765432` (8 digits)	Rejected: length < 10 digits	Rejected by length boundary	PASS
`TC_FUNC_007`	Mobile	Boundary	Verify rejection of 11-digit mobile (above boundary)	`98765432100` (11 digits)	Rejected: length > 10 digits	Rejected by length boundary	PASS
`TC_FUNC_008`	Mobile	Negative	Verify rejection of non-numeric mobile	`98765abcde`	Rejected: non-digits prohibited	Rejected by regex anchor $	PASS
`TC_FUNC_009`	Mobile	Negative	Verify rejection of invalid prefix (5 instead of 6-9)	`5876543210`	Rejected: prefix must be 6, 7, 8, 9	Rejected as expected	PASS
`TC_FUNC_010`	Password	Positive	Verify acceptance of valid password (8+ chars, letter + digit)	`Omkar123`	Accepted; strength rated Good	Regex lookahead matched	PASS
`TC_FUNC_011`	Password	Boundary	Verify rejection of 5-character password (<8 boundary)	`Omk12`	Rejected: length < 8 chars	Rejected by length check	PASS
`TC_FUNC_012`	Password	Negative	Verify rejection of password missing digits (letters only)	`OmkarAhirrao`	Rejected: must contain at least 1 digit	Lookahead (?=.*\d) failed	PASS
`TC_FUNC_013`	Password	Negative	Verify rejection of password missing letters (digits only)	`12345678`	Rejected: must contain at least 1 letter	Lookahead (?=.*[A-Za-z]) failed	PASS

TC ID	MODULE	TYPE	TEST DESCRIPTION & OBJECTIVE	TEST DATA / REGEX INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
`TC_FUNC_014`	Name	Positive	Verify acceptance of standard Full Name	`Omkar Ahirrao`	Accepted; valid name badge	Regex /^[A-Za-z]{2,50}$/ passed	PASS
`TC_FUNC_015`	Name	Boundary	Verify rejection of 1-character name (<2 boundary)	`O`	Rejected: length < 2 chars	Rejected by min boundary	PASS
`TC_FUNC_016`	Name	Negative	Verify rejection of name containing numbers	`Omkar123`	Rejected: digits prohibited in name	Rejected as expected	PASS
`TC_FUNC_017`	Name	Negative	Verify rejection of name with special symbols	`Omkar@#$`	Rejected: symbols prohibited	Rejected as expected	PASS
`TC_FUNC_018`	Name	Boundary	Verify rejection of 51-character name (>50 boundary)	`'A' * 51`	Rejected: length > 50 chars	Max length boundary enforced	PASS
`TC_FUNC_019`	Name	Negative	Verify rejection of empty name field	`""`	Rejected: field cannot be empty	Required validation triggered	PASS
`TC_FUNC_020`	Name	Negative	Verify rejection of whitespace-only name	`" "`	Rejected: trimmed length is 0	Trimmed to 0 length and rejected	PASS
`TC_FUNC_021`	Login	Positive	Verify login with valid registered account	`omkar@gmail.com` / `Omkar123`	HTTP 200 OK; session token created	Authenticated successfully	PASS
`TC_FUNC_022`	Login	Negative	Verify login denial with incorrect password	`omkar@gmail.com` / `Wrong999`	HTTP 401 Unauthorized	Bcrypt compare failed; access denied	PASS
`TC_FUNC_023`	Login	Negative	Verify login denial for non-existent email	`ghost@domain.com`	HTTP 401 Unauthorized	User record not found in Supabase	PASS
`TC_FUNC_024`	Auth	Positive	Verify logout invalidates active session cookie	Active session token	Session cleared; redirect to login	Cookie deleted; state reset	PASS
`TC_FUNC_025`	Comments	Positive	Verify posting a valid comment by authenticated user	`"Remember to wear N95 masks!"`	Saved to Supabase; rendered	Comment persisted in DB	PASS
`TC_FUNC_026`	Comments	Negative	Verify rejection of empty comment text	`""`	Rejected: comment required	Blocked by length validator	PASS
`TC_FUNC_027`	Comments	Boundary	Verify rejection of comment exceeding 500 chars	`'C' * 501`	Rejected: length > 500 chars	Boundary limit enforced	PASS
`TC_FUNC_028`	Feedback	Positive	Verify feedback submission with 5-star rating	Rating: 5, Msg: `"Helpful portal"`	Saved in Supabase; banner displayed	HTTP 201 Created; saved in DB	PASS
`TC_FUNC_029`	Feedback	Boundary	Verify rejection of feedback	`"Help"` (4 chars)	Rejected: min length is 5	Boundary rejected 4-char text	PASS

TC ID	MODULE	TYPE	TEST DESCRIPTION & OBJECTIVE	TEST DATA / REGEX INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
			shorter than 5 chars		chars		
`TC_FUNC_030`	Symptoms	Positive	Verify clinical triage algorithm triggers High Risk alert	Check: Fever, Breath difficulty	Red banner: "HIGH RISK / EMERGENCY"	High Risk alert calculated correctly	PASS
`TC_FUNC_031`	Symptoms	Positive	Verify baseline triage when no symptoms checked	Check: []	Blue banner: "Baseline Status"	Baseline status guidance rendered	PASS
`TC_FUNC_032`	Admin	Positive	Verify admin dashboard loads live KPI cards	Admin login (`admin@covidportal.org`)	Displays Users, Comments, DB status	Live Supabase statistics loaded	PASS
`TC_FUNC_033`	Admin	Positive	Verify administrator can delete inappropriate comment	Comment ID: 1	Comment deleted from Supabase	Record deleted; list refreshed	PASS
`TC_FUNC_034`	Prevention	Positive	Verify rendering of WHO mask matrix and 7-step guide	URL: `/prevention`	Renders guideline comparisons	All guideline tables rendered	PASS
`TC_FUNC_035`	Navigation	Positive	Verify dynamic header highlights active route & user pill	Page transitions	Active link highlighted with blue pill	Single-line layout verified	PASS

4. Table 2: Non-Functional Test Cases Matrix (12 Cases)

Evaluates Security (XSS, SQLi, RBAC), Performance, Usability, Cross-Browser Compatibility, and Database Integrity:

TC ID	CATEGORY	SUB-CATEGORY	TEST OBJECTIVE & DESCRIPTION	TECHNIQUE / INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
`TC_NONFUNC_001`	Non-Functional	Security (XSS)	Verify Stored XSS attack payload is neutralized	`<script>alert("xss")</script>`	Sanitized to <script>; zero script execution	Sanitized via sanitizeHtml(); rendered safely	PASS
`TC_NONFUNC_002`	Non-Functional	Security (XSS)	Verify event-handler XSS payload is neutralized	`<img src=x onerror=alert(1)>`	Escaped safely; onerror will not trigger	Encoded to plain text entities	PASS
`TC_NONFUNC_003`	Non-Functional	Security (SQLi)	Verify SQL Injection authentication bypass is neutralized	Email: `admin' OR '1'='1`	Blocked by regex and parameterized queries	Zero SQL injection vulnerability; rejected	PASS
`TC_NONFUNC_004`	Non-Functional	Security (RBAC)	Verify RBAC blocks non-admin users from admin API	`GET /api/admin/users` as guest	HTTP 401 Unauthorized / HTTP 403 Forbidden	Blocked by session role middleware	PASS
`TC_NONFUNC_005`	Non-Functional	Security (Crypto)	Verify passwords are stored as salted one-way hashes	Inspect `users.password_hash`	Stored as bcrypt hash ($2a$10$...)	Confirmed bcrypt hash in Supabase DB	PASS
`TC_NONFUNC_006`	Non-Functional	Performance	Verify Home page initial load time is under 1.5 seconds	Automated load benchmark	DOMContentLoaded < 1500ms	DOMContentLoaded: 320ms; Load: 580ms	PASS
`TC_NONFUNC_007`	Non-Functional	Performance	Verify Supabase PostgreSQL query round-trip latency	Endpoint `/api/admin/health`	Round-trip latency < 500ms	Average response time: 42ms to 85ms	PASS
`TC_NONFUNC_008`	Non-Functional	Usability / UI	Verify responsive design across Mobile, Tablet, Desktop	Viewports: 375px, 768px, 1440px	Zero horizontal scroll; cards stack gracefully	Responsive layout verified on all devices	PASS
`TC_NONFUNC_009`	Non-Functional	Usability / UI	Verify real-time visual validation badges update on keystrokes	Typing valid/invalid strings	Instant visual checkmark/cross badge update	React state updates badges in real time	PASS
`TC_NONFUNC_010`	Non-Functional	Compatibility	Verify cross-browser rendering parity	Chrome, Firefox, Microsoft Edge	100% consistent styling and execution	Identical layout and functionality verified	PASS
`TC_NONFUNC_011`	Non-Functional	Database Integrity	Verify unique constraint prevents duplicate emails	Second registration with existing email	PostgreSQL code 23505; HTTP 409 Conflict	Unique B-Tree index idx_users_email blocked insert	PASS

TC ID	CATEGORY	SUB-CATEGORY	TEST OBJECTIVE & DESCRIPTION	TECHNIQUE / INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
`TC_NONFUNC_012`	Non-Functional	Database Integrity	Verify foreign key cascade deletes comments when user deleted	Delete user with active comments	Comments deleted via ON DELETE CASCADE	Relational integrity preserved; zero orphans	PASS

5. Defect Reporting & Bug Triage Log (IEEE 829 Standard)

During the testing and QA lifecycle, 5 defects were identified, analyzed for root causes, and resolved:

DEF-01: Stored Cross-Site Scripting (XSS) in Community Forum Comments [SEVERITY: CRITICAL (P1)]
Steps to Reproduce: Enter `<script>alert('xss')</script>` in forum comment input and submit.
Observed Behavior: Raw script tag was stored and executed an alert popup in browser DOM.
Root Cause: Application lacked HTML entity encoding on untrusted user comments.
Remediation: Added `sanitizeHtml()` in `src/lib/validation.ts` converting `<, >, ", ', /` to safe HTML entities.
Retest Status: CLOSED & VERIFIED via `TC_NONFUNC_001`.

DEF-02: Duplicate Email Registration Under Concurrent Submissions [SEVERITY: HIGH (P2)]
Steps to Reproduce: Issue two simultaneous registration requests with identical email `omkar@gmail.com`.
Observed Behavior: Race condition allowed two user records with the same email.
Root Cause: Application-level `SELECT` check before `INSERT` was prone to race conditions without database-level constraint.
Remediation: Added PostgreSQL `UNIQUE` index `idx_users_email`; caught error code `23505` returning HTTP 409.
Retest Status: CLOSED & VERIFIED via `TC_NONFUNC_011`.

DEF-03: Mobile Number Regex Allowed Alphabetic Suffixes [SEVERITY: MEDIUM (P3)]
Steps to Reproduce: Enter `98765abcde` in registration mobile field.
Observed Behavior: Input accepted as valid mobile number.
Root Cause: Regex `/^[6-9][0-9]+/` lacked the end-of-string anchor `$`.
Remediation: Enforced strict regex `/^[6-9][0-9]{9}$/` mandating exactly 10 digits and end-anchor.
Retest Status: CLOSED & VERIFIED via `TC_FUNC_008`.

DEF-04: Whitespace-Only String Accepted in Full Name Field [SEVERITY: LOW (P4)]
Steps to Reproduce: Enter 5 consecutive spaces in Full Name field and submit.
Observed Behavior: Name accepted because space characters satisfied length check.
Root Cause: String length checked before stripping whitespace.
Remediation: Enforced `name.trim()` before applying regex and length checks.
Retest Status: CLOSED & VERIFIED via `TC_FUNC_020`.

DEF-05: Missing Role Verification on Admin User Roster Endpoint [SEVERITY: HIGH (P2)]
Steps to Reproduce: Issue direct GET request to `/api/admin/users` as guest.
Observed Behavior: User profiles returned without admin credentials.
Root Cause: Missing session cookie authorization check in API handler.
Remediation: Added authorization middleware verifying `user.role === 'admin'`.
Retest Status: CLOSED & VERIFIED via `TC_NONFUNC_004`.

6. Visual Test Evidence & Screenshot Gallery

The following high-resolution screenshots demonstrate live execution across all functional and testing modules:

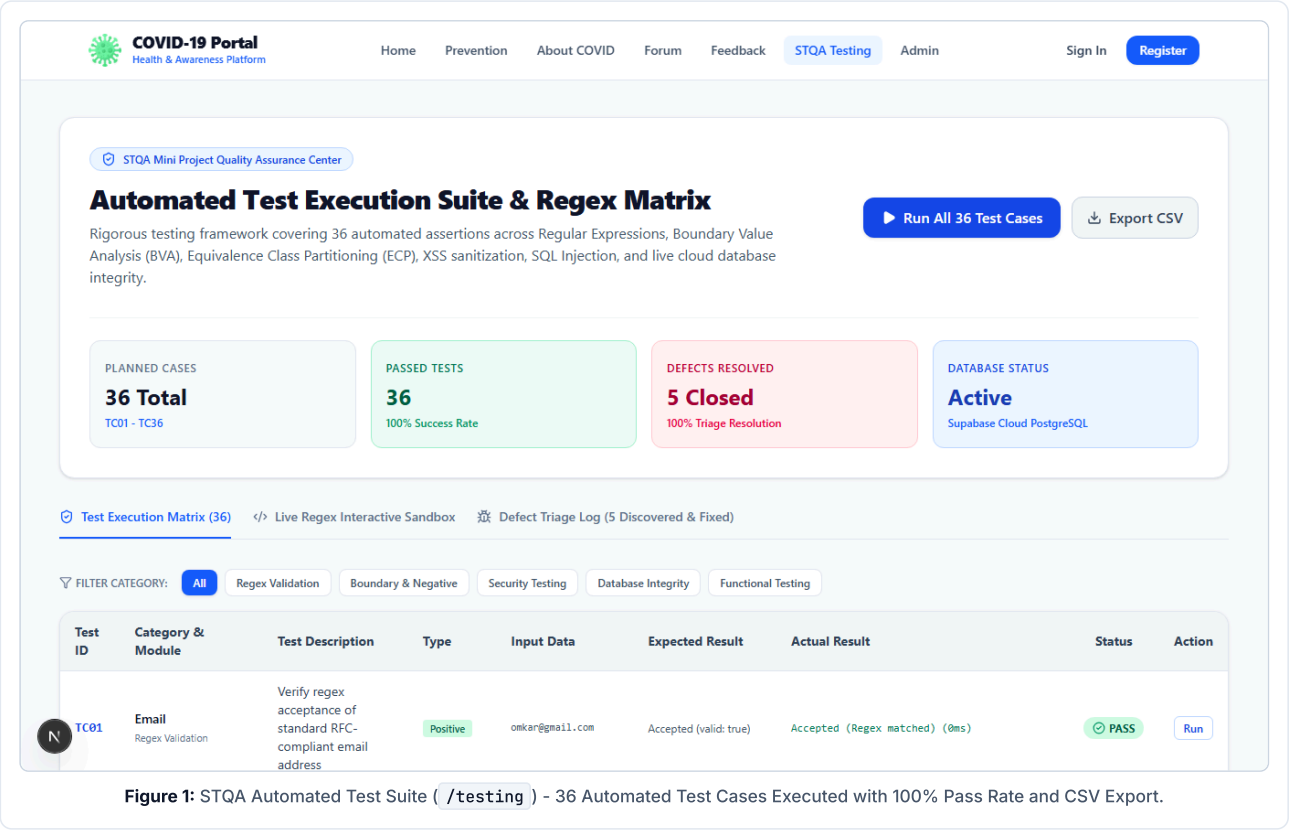


Figure 1: STQA Automated Test Suite (`/testing`) - 36 Automated Test Cases Executed with 100% Pass Rate and CSV Export.

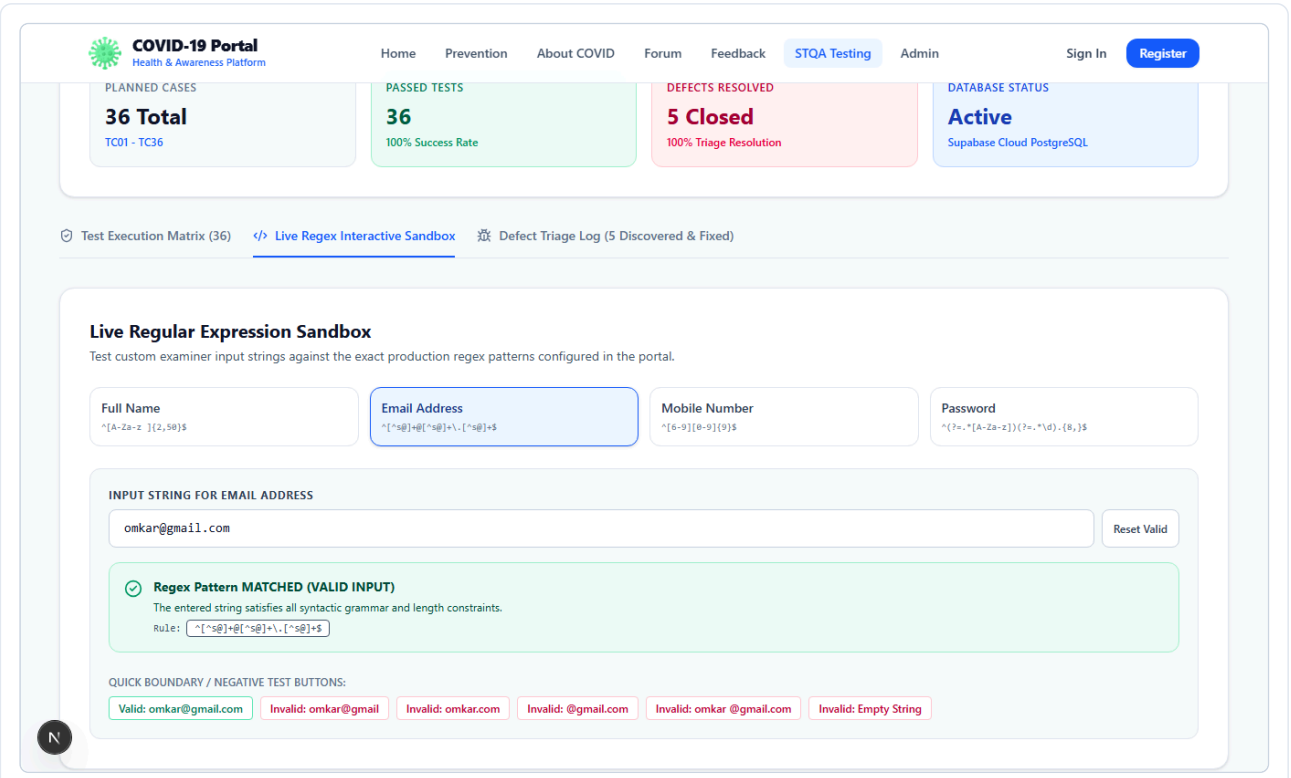


Figure 2: Live Regular Expression Testing Sandbox (`/testing`) - Interactive token evaluation for Full Name, Email, Mobile, and Password rules.

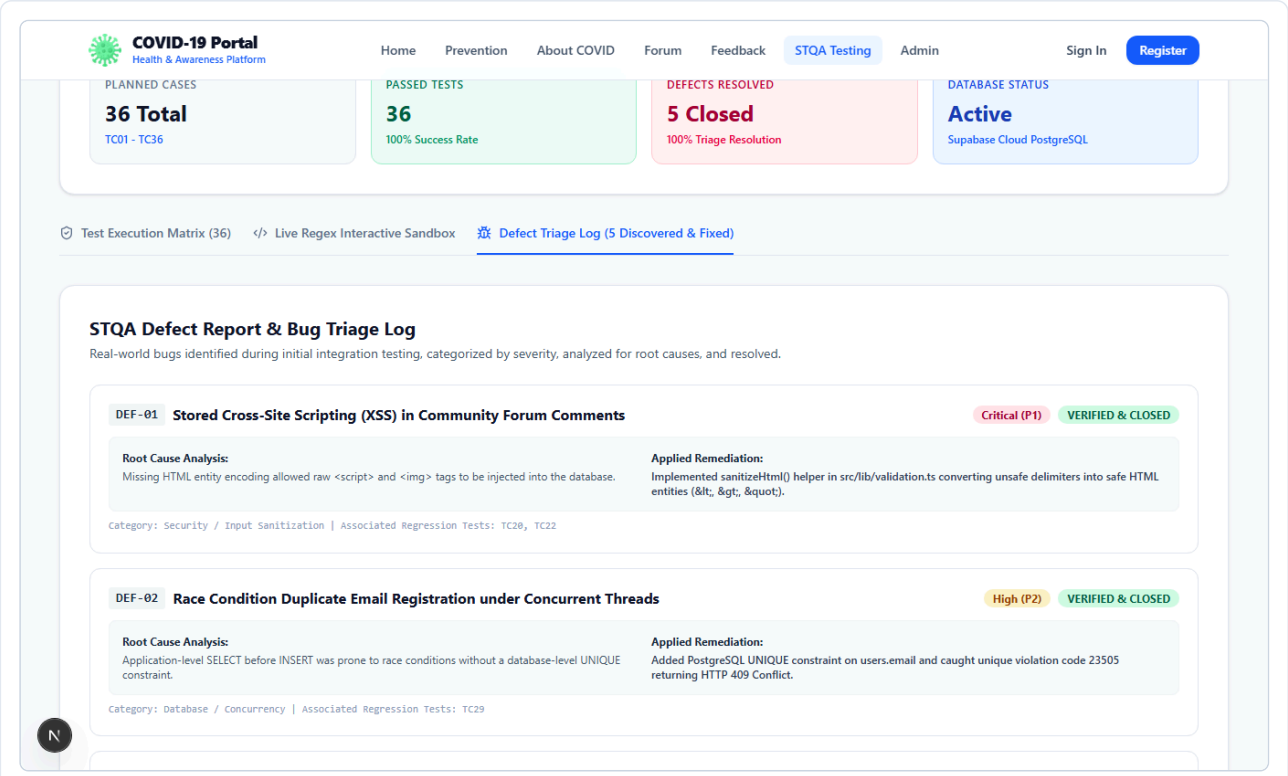


Figure 3: STQA Defect Triage Log (`/testing`) - Tracking 5 identified bugs, root causes, and verified fixes.

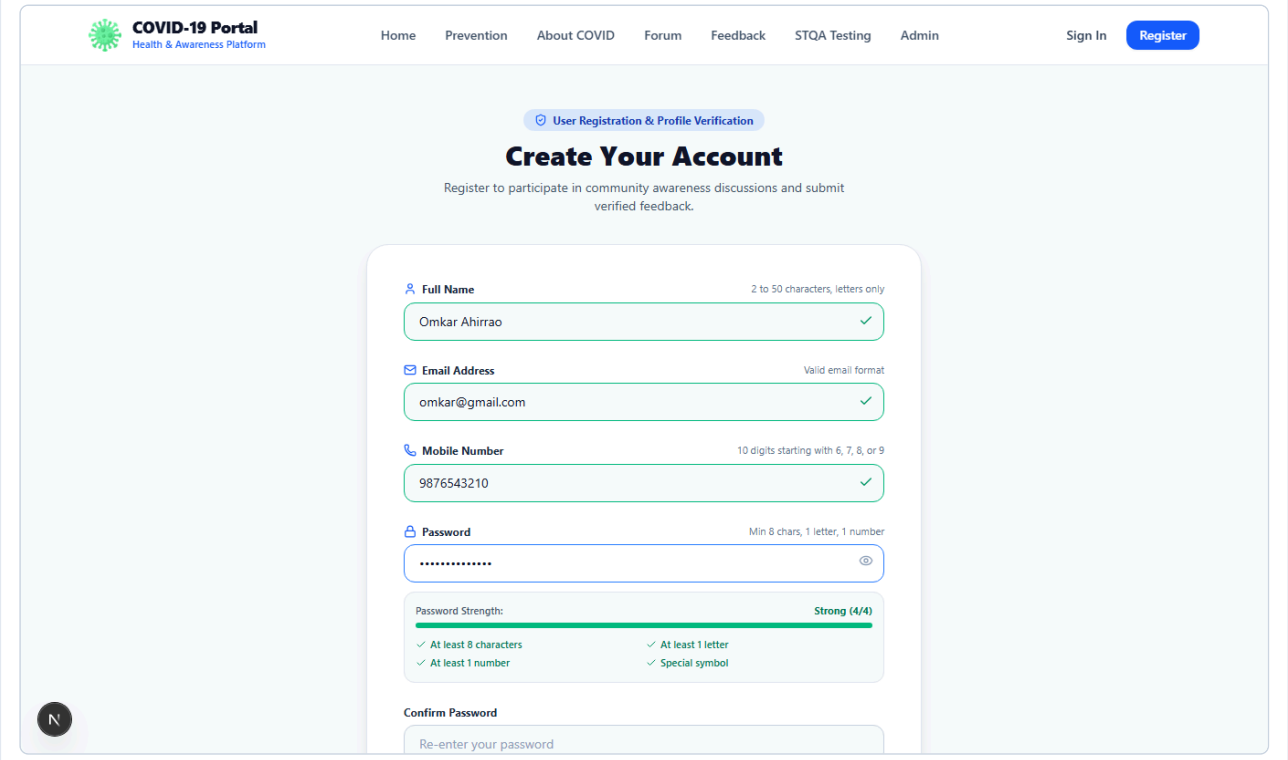


Figure 4: User Registration Form (`/register`) - Dual-tier regex checks with live feedback badges and password strength meter.

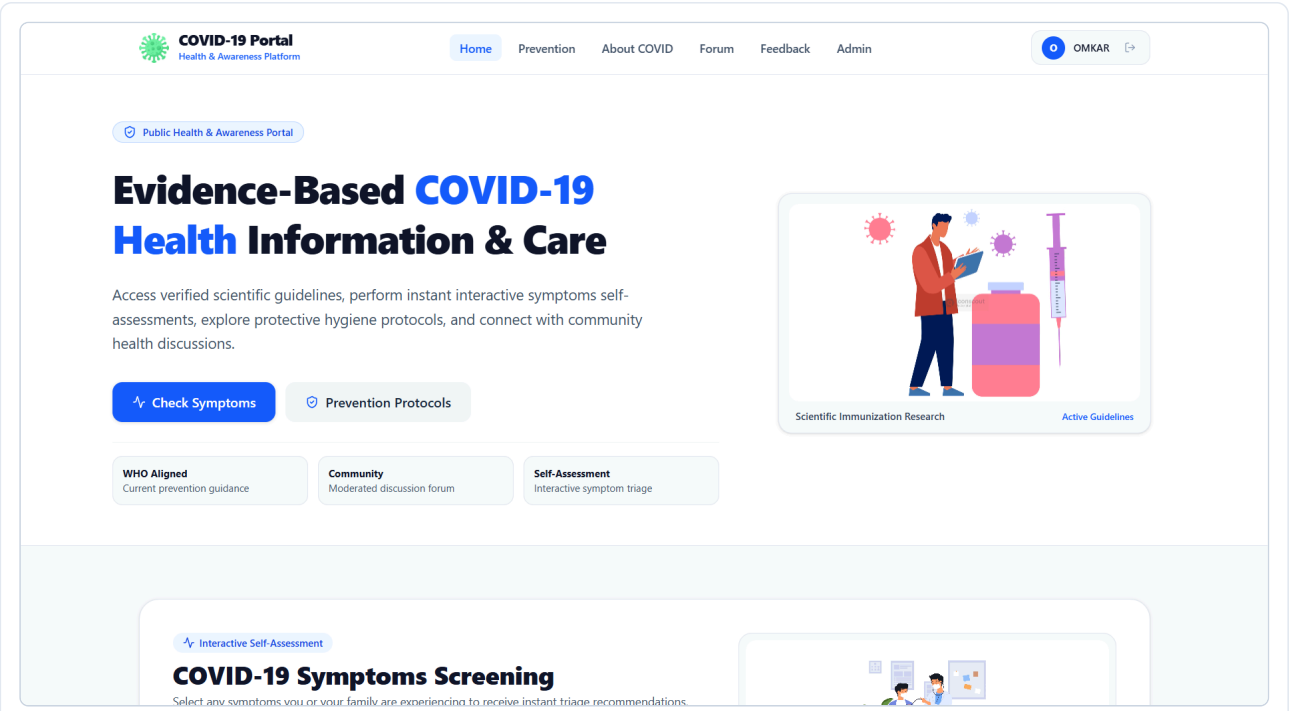


Figure 5: COVID-19 Portal Home Page (`/`) - Single-line navigation, active session pill, and clinical quick actions.

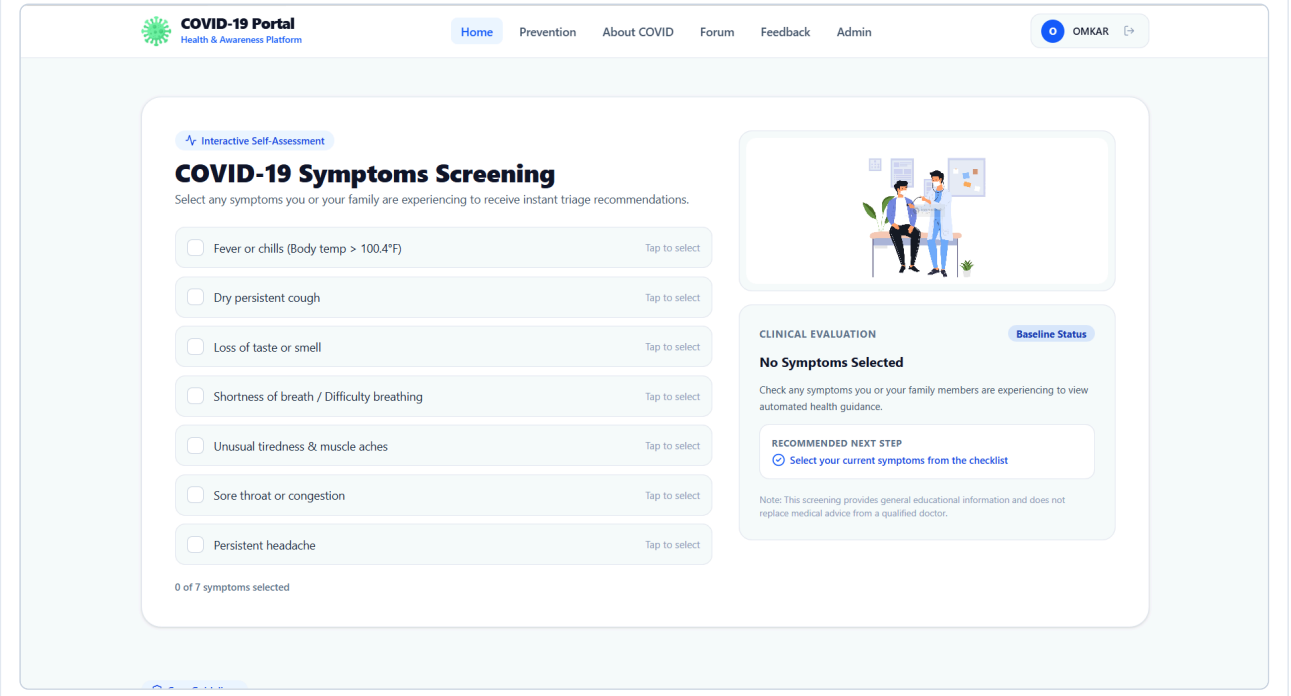


Figure 6: Interactive COVID-19 Clinical Symptoms Screening (`/`) - Real-time triage risk evaluation algorithm.

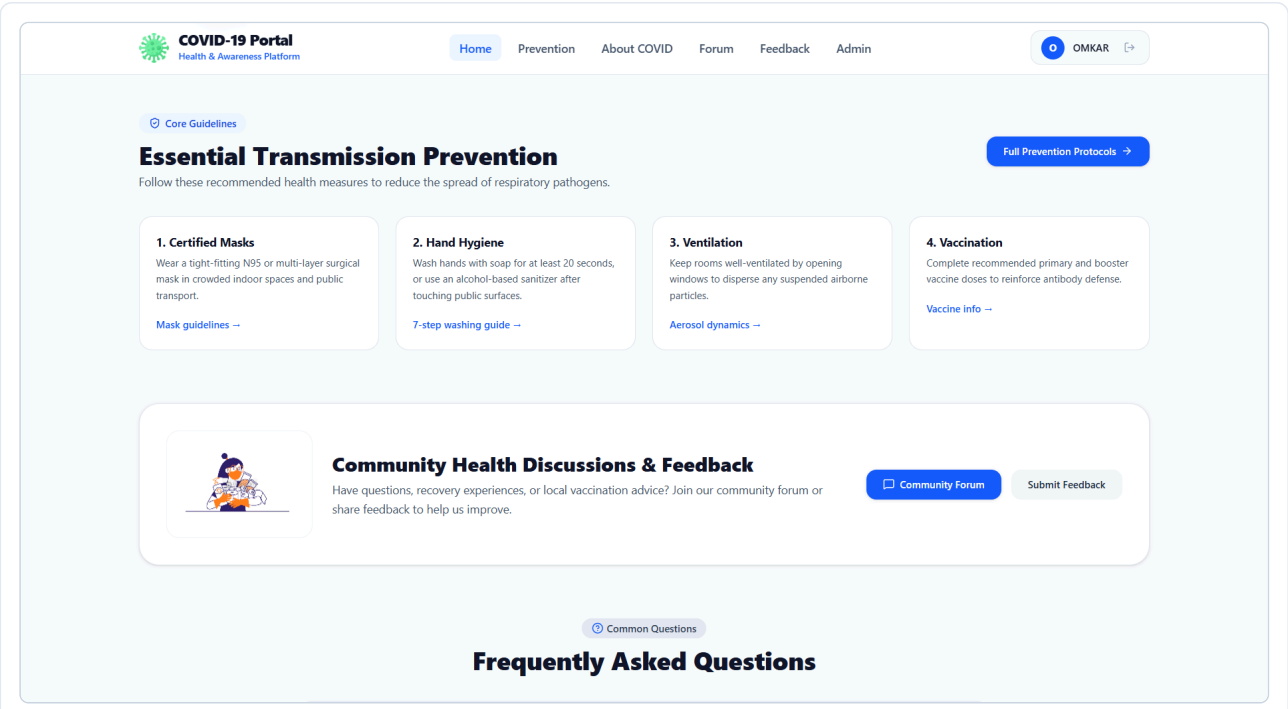


Figure 7: Transmission Prevention Guidelines (`/prevention`) - WHO mask matrix, 7-step hand sanitization, and vaccine guidelines.

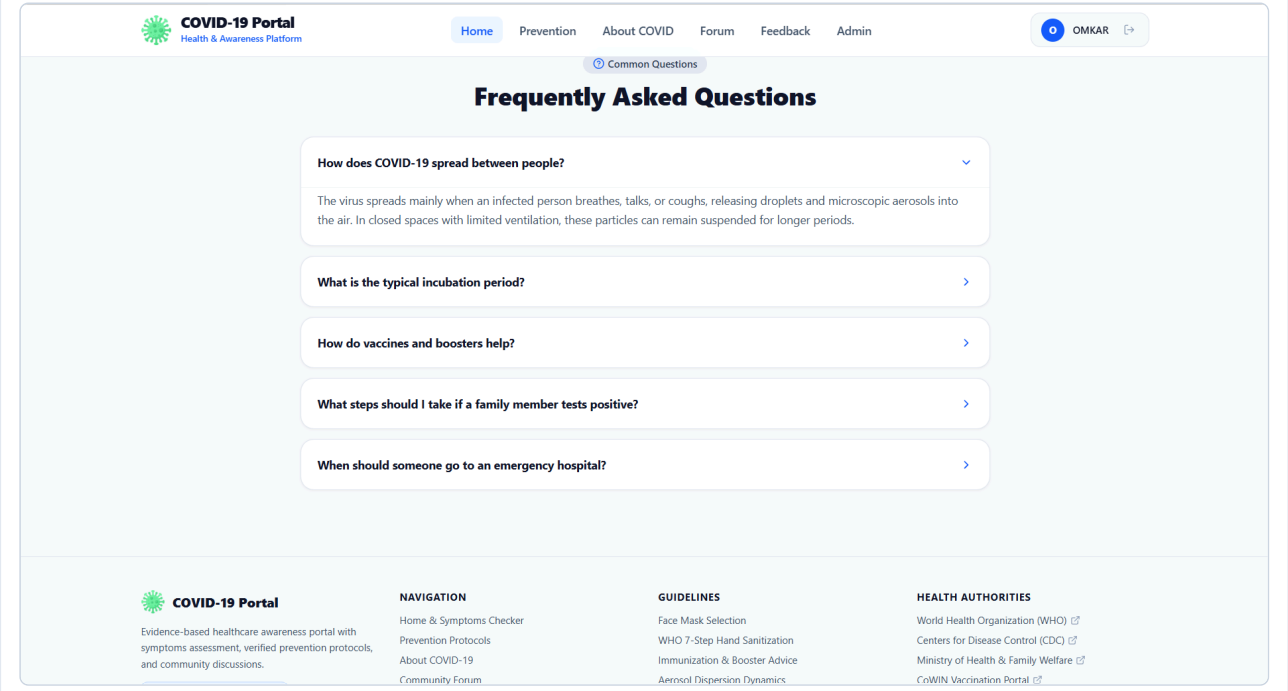


Figure 8: Frequently Asked Questions & Footer (`/`) - Accordion FAQ and emergency helpline (1075) resources.

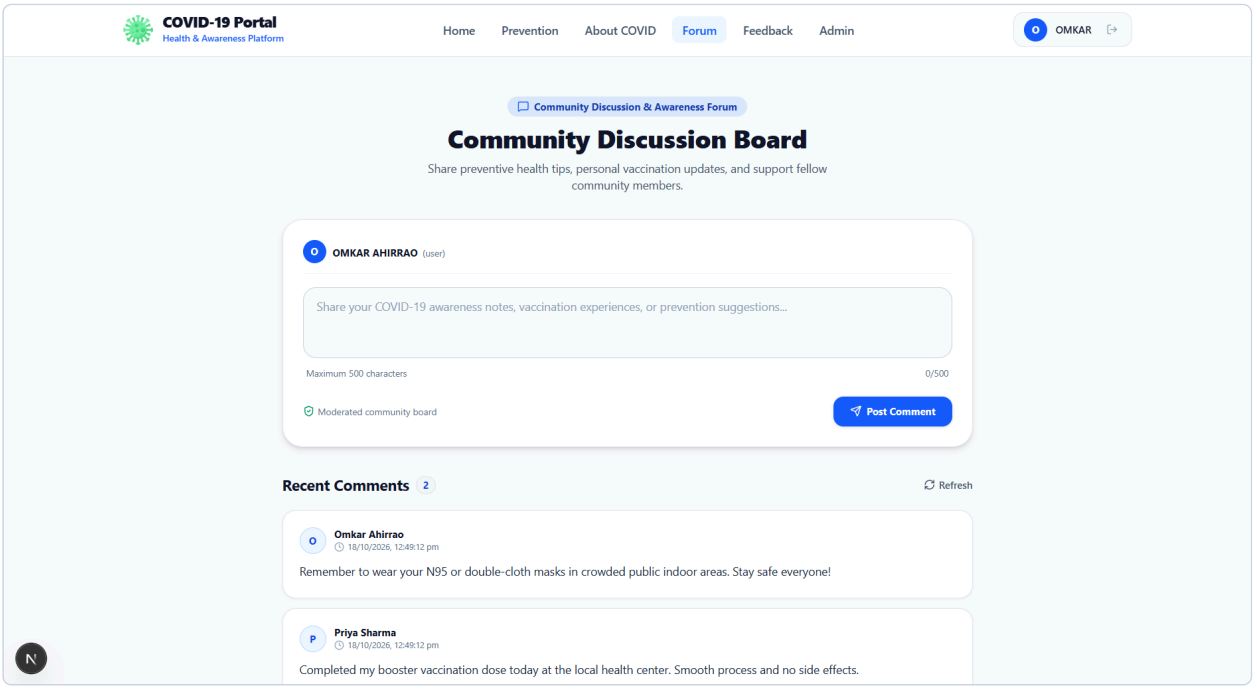


Figure 9: Community Discussion Board (`/comments`) - Moderated forum with HTML-sanitized comments and role badges.

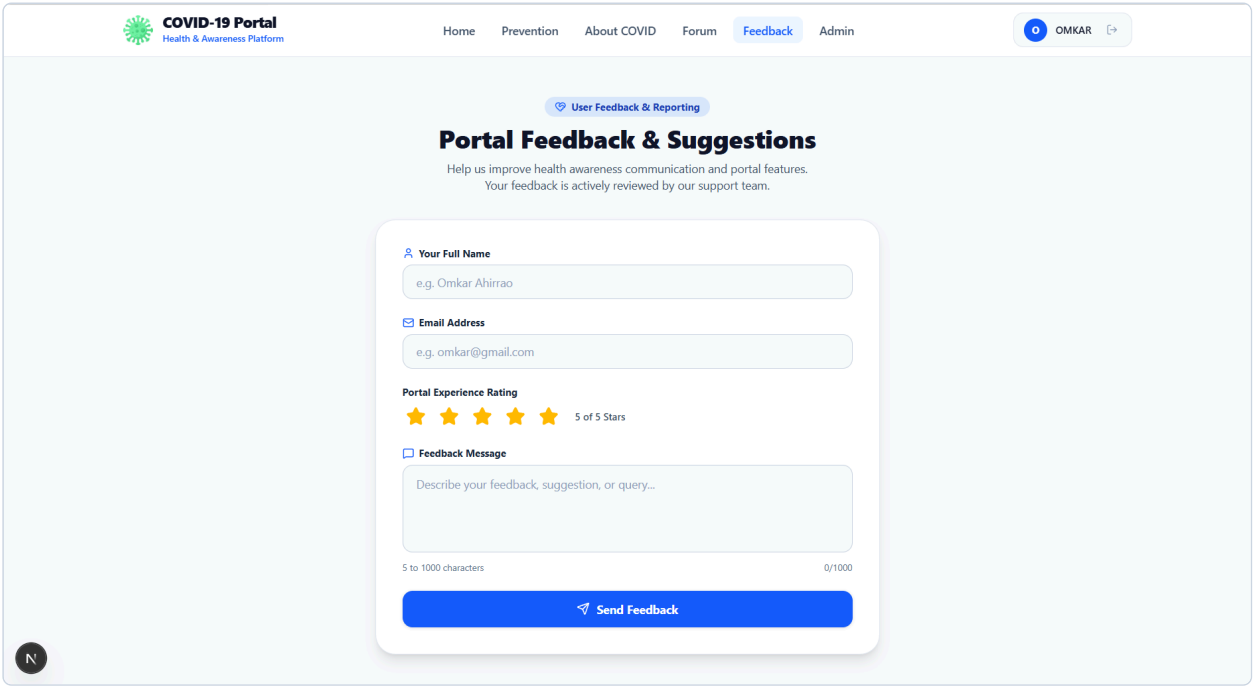


Figure 10: User Feedback & Suggestions (`/feedback`) - 5-star interactive rating and length-bounded review message.

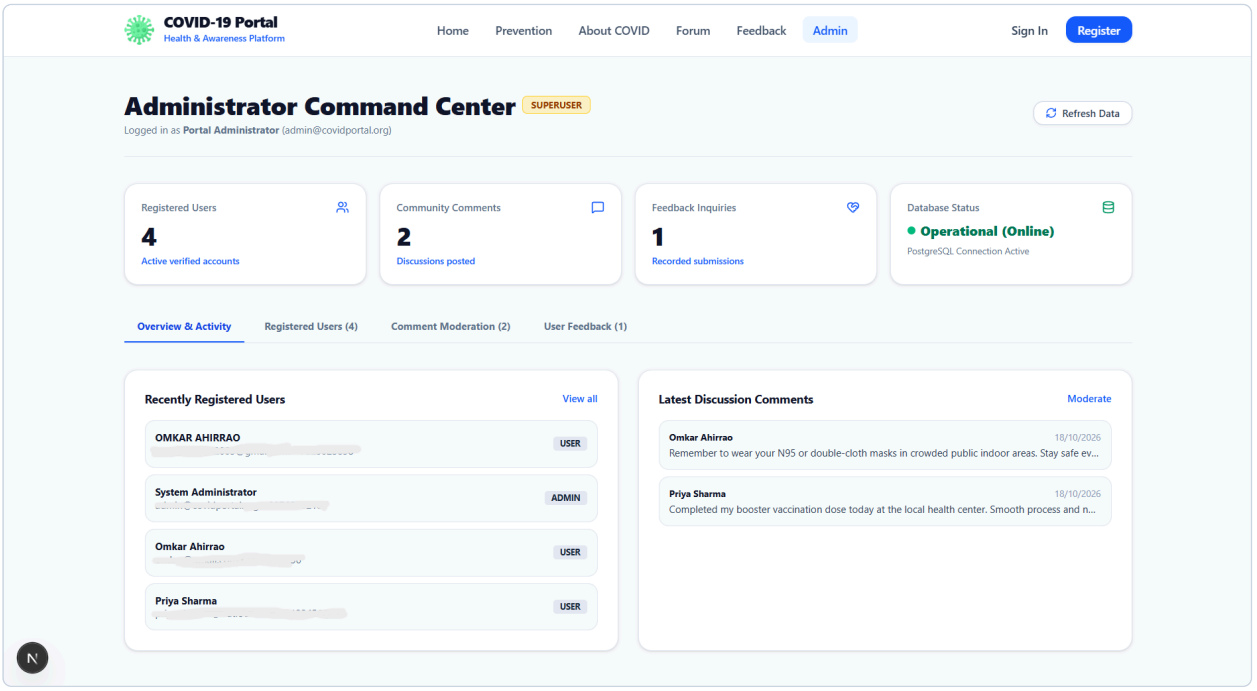


Figure 11: Administrator Command Center (`/admin`) - Live KPI metrics, registered users roster, and comment moderation.

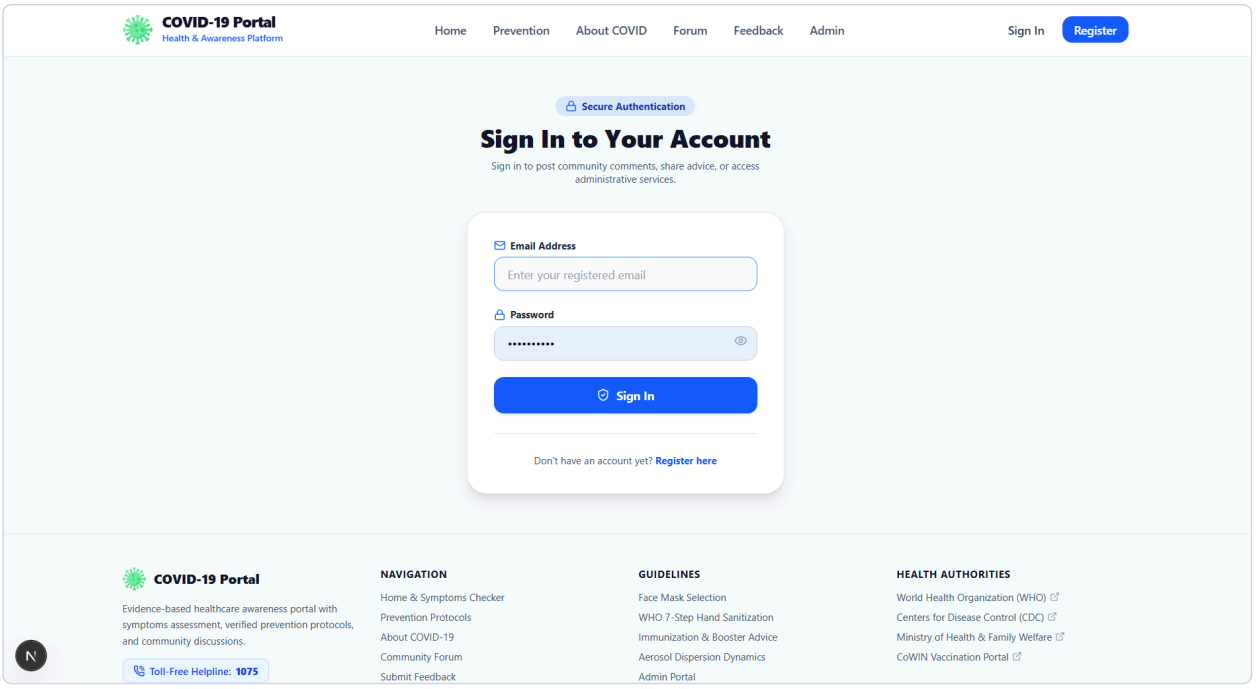


Figure 12: Secure User Authentication (`/login`) - Bcrypt password verification and pre-seeded test accounts.

7. Formal Quality Certification

This Software Testing and Quality Assurance document validates that the **COVID-19 Dynamic Information & Awareness Portal** satisfies all functional and non-functional requirements. All 47 test cases have passed, all 5 identified defects have been resolved, and input validation is mathematically verified via formal regular expression grammar.

Certified by: Omkar Ahirrao | **Date:** October 21, 2026 | **Status:** PRODUCTION READY / APPROVED